Function Context: C1

Function Background

This function comes from Requests' HTTP request-preparation logic. Before a network request is sent, the user-provided URL must be parsed, validated, and normalized.

Input and Output

Inputs: url, the original URL; params, query parameters to append to the URL.

Output: The function does not directly return a result.

Side effect: The processed URL is written to the current request object's self.url attribute.

Exceptions: The function raises an exception when the URL is malformed or lacks required components.

Key Terms

Term	Meaning
URL	The address of a network resource.
scheme	The protocol part of a URL, such as http or https.
host	The hostname in a URL, such as example.com.
port	The port number in a URL.
path	The path identifying a resource within a URL.
query	The query parameters following ? in a URL.
fragment	The fragment identifier following # in a URL.
IDNA	An encoding mechanism for internationalized domain names.
netloc	The URL component containing authentication information, host, and port.

Helper Variables and Functions

Name	Brief Description
parse_url(url)	Splits a URL into its components.
_get_idna_encoded_host(host)	Encodes an internationalized hostname.
_encode_params(params)	Encodes parameters as a query string.
urlunparse(...)	Reassembles URL components into a string.
requote_uri(...)	Normalizes URI escaping.
to_native_string(...)	Converts input to a string representation suitable for subsequent processing.

Background Not Required

Participants do not need to know the full HTTP specification. They only need to use the code to determine how the URL is processed under different conditions.